



EchoSim

Interactive 3D in-silico echocardiography trainer

Fidelity 5.9 → **7.4** / 10 · v2 state-of-the-art rubric

Benchmark evolution & evidence deck · 2026-07-10

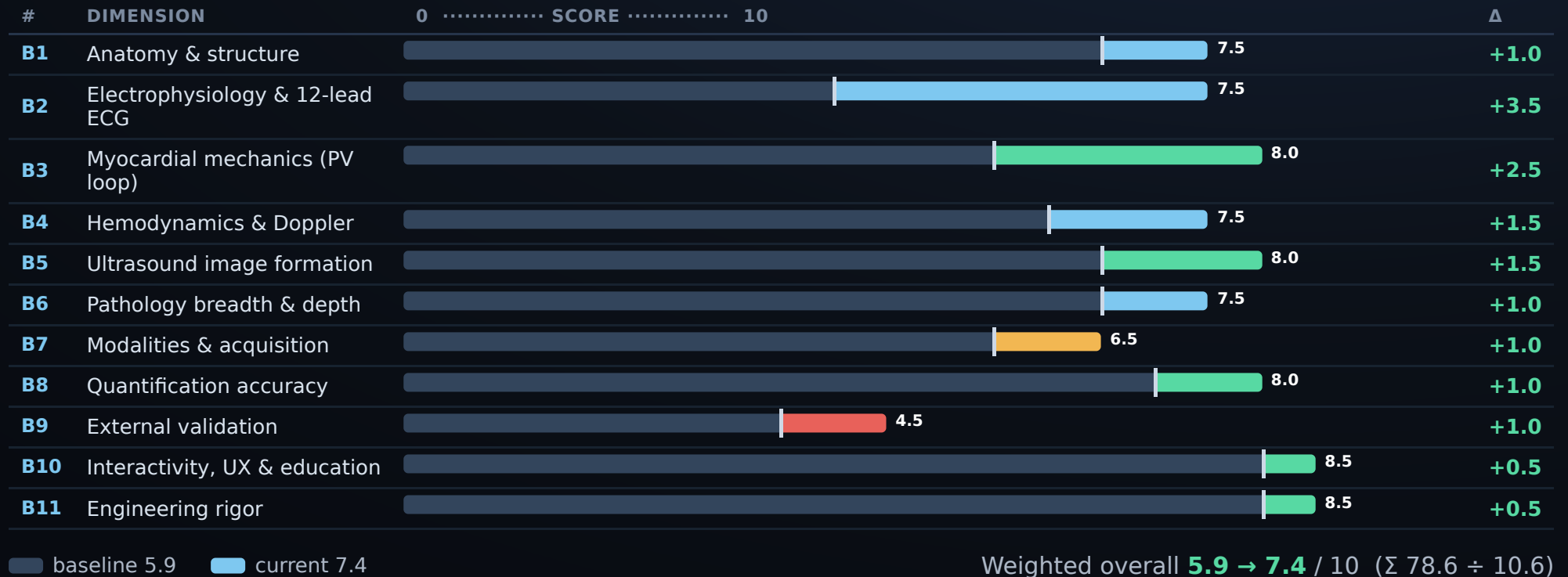
What it is

- A beating 3D heart you scan like a real machine — a moveable ultrasound plane drives a live 2-D echo.
- Physics-grounded: a **12-lead ECG**, a lumped-parameter **PV loop**, continuity **Doppler**, and scatterer-based **B-mode**.
- Standard views (PLAX/PSAX/A4C/A2C/subcostal), 10 graded pathologies, learner measurements, challenge mode.
- Fully offline, no PHI — deployable today as an **education-only** trainer.



Benchmark evolution — baseline → current

Weighted against physics-based computational cardiology & validated commercial simulators (not a teaching-tool self-rating).



Physics-derived 12-lead ECG B2

· 4.0→7.5

- Vectorcardiographic dipole → **inverse-Dower lead field** → standard 12-lead.
- Rhythm library: sinus, AF, brady/tachy, AV block, PVCs, LBBB/RBBB, STEMI, hyperK.
- Electromechanically coupled to the cardiac cycle; **12/12** automated morphology checks pass.

Anterior STEMI — ST-elevation vector projects to the anterior leads.

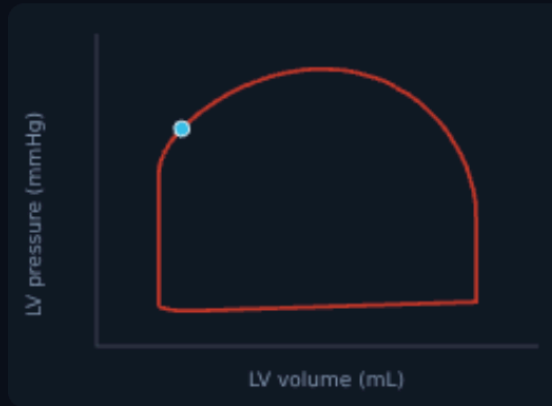
Normal sinus 12-lead: aVR–, normal axis, R-wave progression V1→V6.

PV loop, haemodynamics & image quality

B3/B5 • →7.5

- Lumped-parameter closed-loop circulation → a real LV **pressure-volume loop**; EF/LVIDs fall out of the physics.
- Live **gCNR / CNR / speckle-SNR** self-validation of the scatterer+PSF B-mode.
- Severe AS shown: P_{LV} 149, gradient 76 mmHg, gCNR 0.95.

HAEMODYNAMICS lumped-parameter PV loop



EDV	94 mL	ESV	40 mL
SV	54 mL	EF	57 %
P_{LV} peak	160 mmHg	Aortic	92/65
AV ΔP_{peak}	82 mmHg	Regurg. frac.	—
Est. PASP	—	E/A	1.19
GLS	-22.6 %		

P_{LV}/P_{ao} mmHg · volumes mL. Loop area = stroke work; width = stroke volume.

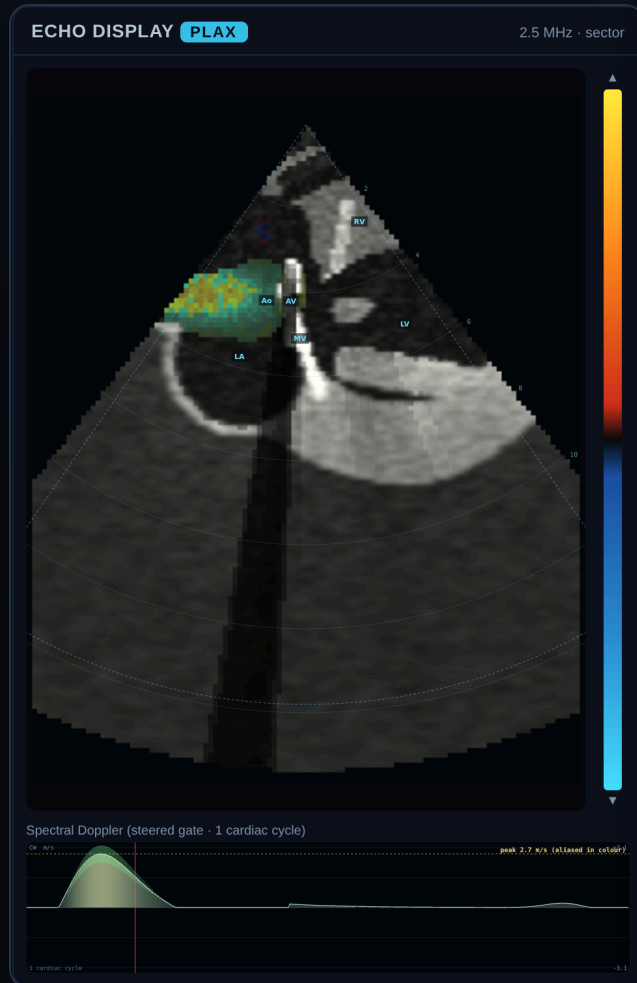
REGIONAL STRAIN GLS bullseye (17-segment)



trative.

Graded pathology & continuity Doppler

B4/B6 → 7.5



Aortic stenosis — jet velocity is continuity-derived ($v=Q/A$); graded mild→severe.



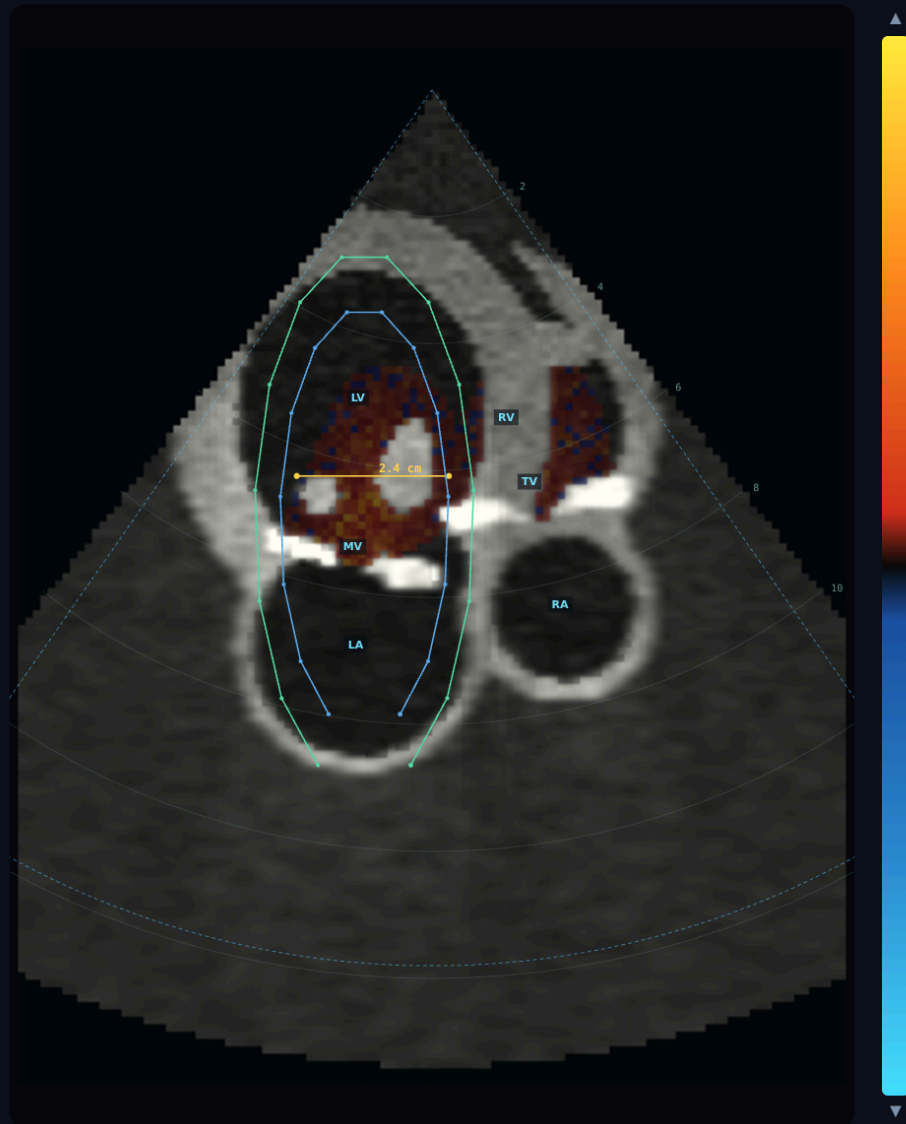
Mitral regurgitation — Bernoulli jet, regurgitant fraction 26/46/65% by grade.



Regional wall-motion abnormality — infarcted septum fails to thicken; EF ~46%.

ECHO DISPLAY **A4C**

2.5 MHz · sector



Spectral Doppler (steered gate · 1 cardiac cycle)



Learner measurement tools **B8**

7.0→8.0

- Click-two-point **calipers** (exact cm) and a traced **Simpson method-of-discs EF**.
- Checked **live against the model's ground truth** — measured 53 %, error -5 % (truth 59), within the ASE $\pm 6\%$ band.
- Disc-summation math verified to **0.0%** vs an analytic solid of revolution.

Honest status & the path beyond 7.4

Deployable today

An **education-only, non-device** trainer — offline, no PHI, physics-grounded. Verified at **7.4/10**.

What code cannot close

The 9-10 band & a **clinical-training deployment** claim need **B9 external validation** (expert panel + learning-transfer RCT) and **regulatory/QMS clearance** — human evidence-and-audit processes. The protocol, instrument & harness are shipped in the repo (*VALIDATION.md*, *COMMERCIAL_READINESS.md*, *ROADMAP_TO_9.9.md*).